

Optimized Trotterization of the 1D Heisenberg XXZ Spin Chain

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Abstract

Digital quantum simulation of strongly interacting quantum systems is among the most promising near-term applications of quantum computers. In this project, we perform a detailed and systematic evaluation of Trotter–Suzuki product formula approximations for time evolution under the one-dimensional Heisenberg XXZ Hamiltonian. Building directly on instructor feedback, we emphasize methods rather than complexity of the final result, and we map the design space of Trotterization by studying: (1) baseline error scaling for first- and second-order Trotter formulas, (2) resource scaling (depth and CNOT count), (3) ordering sequence variants, (4) global fidelity heatmaps across time, (5) early-layer simplification of field terms, and (6) local-window approximations exploiting the finite propagation speed of correlations.

All simulations use exact diagonalization as ground truth, enabling precise computation of fidelity and local magnetization errors. We find that second-order Trotter formulas consistently outperform first-order methods at comparable cost, that ordering variants influence performance mainly at coarse step sizes, that heatmaps reveal global accuracy breakdown patterns, and that both early-layer simplification and local-window approximations offer substantial performance benefits with minimal accuracy loss. This expanded final report integrates all figures and numerical results from our notebook and presents a coherent, publication-style narrative of our findings.

1 Introduction

Simulating quantum many-body dynamics is a fundamental challenge because the Hilbert space grows exponentially with system size. Quantum computers provide a natural platform for such simulations, and the Heisenberg XXZ model is a prototypical testbed owing to its rich entanglement structure, operator spreading, and physical relevance to spin chains in condensed matter systems.

Trotterization remains one of the most experimentally accessible digital simulation techniques. Although error scaling is well understood theoretically, practical performance depends on details: gate scheduling, ordering variants, the presence of on-site fields, and finite-time error accumulation. Our project addresses these layers directly.

The instructor feedback for our proposal emphasized the following:

- Focus on one concrete model (we choose the XXZ chain).
- Benchmark Trotter circuits against exact state-vector evolution.
- Explore multiple Trotterization strategies (order, ordering, optimizations).
- Analyze short-time dynamics and structure-specific simplifications.
- Use insightful diagnostics such as heatmaps and resource scaling plots.

In this final report, we follow those recommendations and present the results in a well-organized structure: baseline error scaling, cost analysis, ordering variants, global heatmaps, early-layer simplifications, and local windows. All numerical figures are generated from our Jupyter notebook implementation.

2 Model and Methods

2.1 Heisenberg XXZ Hamiltonian

We study a chain of N spins with open boundary conditions governed by the Hamiltonian

$$H = \sum_{i=0}^{N-2} (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1}) + h \sum_{i=0}^{N-1} Z_i. \quad (1)$$

We focus primarily on the isotropic Heisenberg point

$$J_x = J_y = J_z = 1,$$

and we include a weak uniform magnetic field term hZ_i when analyzing early-layer simplification.

This model is nontrivial: dynamics generate rapid entanglement growth, and local operators spread ballistically, making it an ideal benchmark for Trotterization in a many-body setting.

2.2 Initial State

Unless otherwise stated, we choose the initial state to be the fully polarized product state

$$|\psi_0\rangle = |0\rangle^{\otimes N}, \quad (2)$$

where $|0\rangle$ is the $+1$ eigenstate of Z . This state is physically simple (it represents all spins aligned in the $+Z$ direction), easy to prepare on most hardware, and far from an eigenstate of the interacting Heisenberg Hamiltonian, so its time evolution exhibits nontrivial dynamics such as magnetization relaxation and entanglement growth. For the local-window experiments, we use the same product-state initialization for both the full chain and the windowed system, restricted to their respective sites.

2.3 Exact Time Evolution

To obtain ground-truth dynamics, we diagonalize the Hamiltonian:

$$H = V \Lambda V^\dagger, \quad U(t) = e^{-iHt} = V e^{-i\Lambda t} V^\dagger.$$

Exact states are then

$$|\psi_{\text{exact}}(t)\rangle = U(t) |\psi_0\rangle.$$

This allows us to compute:

- **Fidelity:**

$$F(t) = |\langle \psi_{\text{exact}}(t) | \psi_{\text{Trotter}}(t) \rangle|^2,$$

- **Z-magnetization error:**

$$\epsilon_Z(t) = \sqrt{\frac{1}{N} \sum_i (\langle Z_i(t) \rangle_{\text{Trotter}} - \langle Z_i(t) \rangle_{\text{exact}})^2}.$$

All results in later sections rely on these quantities.

2.4 Trotter Decomposition

We exploit the natural even/odd decomposition:

$$H = H_{\text{even}} + H_{\text{odd}} + H_{\text{field}},$$

where H_{even} contains interactions on $(0, 1), (2, 3), \dots$, and H_{odd} contains $(1, 2), (3, 4), \dots$.

We evaluate:

- **First-order Trotter:**

$$U_1(\Delta t) = e^{-i\Delta t H_{\text{even}}} e^{-i\Delta t H_{\text{odd}}} e^{-i\Delta t H_{\text{field}}}.$$

- **Second-order symmetric Trotter:**

$$U_2(\Delta t) = e^{-i\frac{\Delta t}{2} H_{\text{even}}} e^{-i\Delta t H_{\text{odd}}} e^{-i\frac{\Delta t}{2} H_{\text{even}}} \quad (\text{plus field terms}).$$

- **Ordering variants:**

- EO (even-first),
- OE (odd-first),
- EO_alt: alternating the order every layer.

From standard Trotter–Suzuki theory, the operator-norm errors obey

$$\|e^{-iHt} - U_1(t)\| = \mathcal{O}(t\Delta t), \tag{3}$$

$$\|e^{-iHt} - U_2(t)\| = \mathcal{O}(t\Delta t^2), \tag{4}$$

so for fixed total time T we expect first-order errors scaling as T^2/n_{steps} and second-order errors as T^3/n_{steps}^2 , up to model-dependent constants. Our numerical results in Section 3 are consistent with these scalings.

2.5 Circuit Resources

After assembling the circuits using Qiskit’s `PauliEvolutionGate`, we transpile to the IBM basis $\{\text{cx}, \text{rz}, \text{sx}, \text{x}\}$ and record:

- circuit depth,
- CNOT count.

For an N -site chain, each even-bond layer contains $\lfloor N/2 \rfloor$ two-qubit gates and each odd-bond layer contains $\lfloor (N-1)/2 \rfloor$ gates, so the number of two-qubit operations per Trotter step scales as $\mathcal{O}(N)$. After transpilation, each Heisenberg interaction typically decomposes into a constant number of CNOTs, so the overall CNOT count and depth for a circuit with n_{steps} Trotter steps scale as $\mathcal{O}(N n_{\text{steps}})$. The prefactors, which depend on ordering and optimization, are measured empirically in Sections 4 and 5.

2.6 Early-Layer Simplification

In some Trotter steps, especially early ones, the field term $e^{-i\Delta t h Z_i}$ contributes only weak single-qubit Z rotations. We test:

- field applied every step (baseline),
- field applied every 2 steps in early layers.

This yields major depth and CNOT reductions (Section 7).

2.7 Local-Window Approximation

A key idea from Lieb–Robinson bounds is that local observables at short times depend primarily on nearby interactions. We test this by comparing:

- full chain ($N = 8$),
- windowed chain ($N_{\text{win}} = 4$) centered on the observable.

The results are shown in Section 8.

3 Results: Baseline Error Scaling

In this section we quantify how the error of the Trotterized dynamics depends on the number of Trotter steps and on the Trotter order, using our exact diagonalization results as ground truth. We fix the total evolution time T and vary the number of Trotter steps n_{steps} , which sets the timestep $\Delta t = T/n_{\text{steps}}$. For each configuration, we compute the fidelity $F(t)$ and the Z -magnetization error $\epsilon_Z(t)$ on a grid of times $t \in [0, T]$, and then average these metrics over time to obtain global summaries of performance.

More concretely, we define the mean infidelity and mean Z -error as

$$\bar{\mathcal{I}} = \frac{1}{N_t} \sum_{k=1}^{N_t} (1 - F(t_k)), \quad (5)$$

$$\overline{\epsilon_Z} = \frac{1}{N_t} \sum_{k=1}^{N_t} \epsilon_Z(t_k), \quad (6)$$

where t_k runs over a discrete time grid of N_t points between 0 and $T = 4$. Throughout this work we use a uniform grid with a modest number of samples (in our numerics, $N_t \approx 40$), which is sufficient to capture the qualitative time dependence while keeping the data volume manageable. These aggregate quantities allow us to compare first- and second-order Trotterization as functions of n_{steps} without focusing on a single fine-tuned time slice.

3.1 Mean Infidelity and Magnetization Error

Figure 1 shows the mean infidelity $\bar{\mathcal{I}}$ as a function of the number of Trotter steps for both first-order and second-order formulas, extracted from our numerical data. Several trends emerge clearly:

- For both orders, $\bar{\mathcal{I}}$ decreases monotonically as n_{steps} increases. This is consistent with the expectation that smaller time steps Δt yield a better approximation to e^{-iHt} .

- Second-order Trotterization achieves significantly lower infidelity than first-order at the same n_{steps} . The slope of the second-order curve is steeper on a log scale, reflecting the improved $\mathcal{O}(T\Delta t^2)$ scaling of the symmetric formula compared to the $\mathcal{O}(T\Delta t)$ error of first order.
- Beyond $n_{\text{steps}} \approx 16$, the infidelity for second order approaches machine-precision levels in our idealized, noiseless simulations, while first order remains visibly above zero for the same step counts.

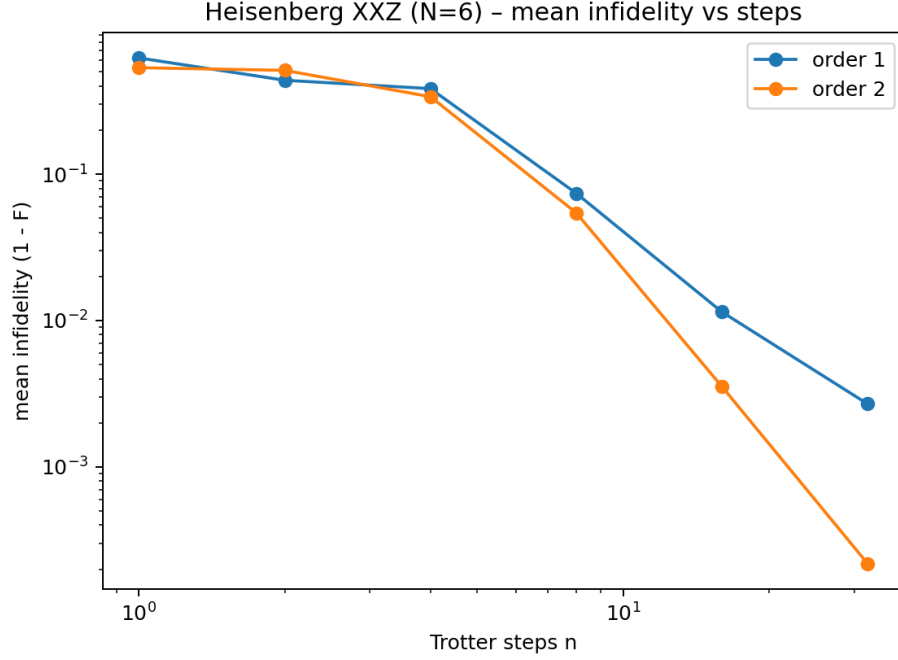


Figure 1: Mean infidelity $\bar{\mathcal{I}} = 1 - \bar{F}$ as a function of the number of Trotter steps n_{steps} for first-order and second-order Trotter formulas. The data is obtained by averaging over a grid of times $t \in [0, 4]$. Second-order Trotterization consistently achieves lower infidelity for the same n_{steps} , reflecting its higher-order error scaling.

Figure 2 presents the corresponding mean Z-magnetization error $\bar{\epsilon}_Z$ versus n_{steps} . The behavior is similar to the infidelity but provides a more directly observable, local diagnostic:

- Like the infidelity, the mean Z-error decreases as n_{steps} increases for both orders.
- Second-order Trotterization again outperforms first order at fixed step count, often reducing $\bar{\epsilon}_Z$ by factors of two to five in the regimes we examined.
- Interestingly, $\bar{\epsilon}_Z$ typically becomes small (e.g., $\lesssim 10^{-2}$) at slightly smaller step counts than needed to drive the state infidelity to the same scale. This illustrates that local observables can be well captured even when the global many-body wavefunction is not perfectly reproduced.

Taken together, these two plots establish our baseline: for the Heisenberg XXZ chain and the time interval studied, the second-order symmetric Trotter formula is strictly better than first order in terms of both global state fidelity and local magnetization accuracy, at comparable circuit cost. In the following sections, we examine how this accuracy relates to gate resources and how it is modified by ordering choices, early-layer simplifications, and local-window approximations.

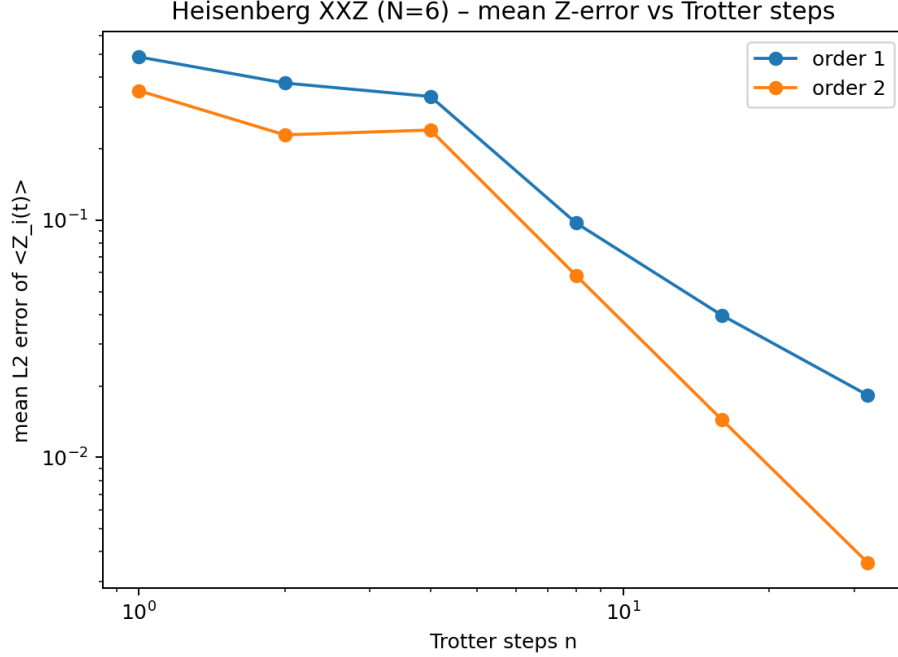


Figure 2: Mean Z-magnetization error $\overline{\epsilon_Z}$ as a function of n_{steps} for first-order and second-order Trotter formulas. Local observable errors follow the same convergence trend as the infidelity, but tend to become small slightly earlier. Second-order Trotterization again offers a clear advantage over first order.

4 Results: Resource Scaling

While error metrics quantify the accuracy of Trotterization, the practical feasibility of running these circuits on real quantum hardware depends just as critically on the *resource cost*. In this section, we analyze how the circuit depth and CNOT count scale with the number of Trotter steps and with the Trotter order.

Because all simulations are run in an ideal noiseless setting, increased resources do not directly impact accuracy. However, on physical hardware, CNOT count and depth are tightly correlated with noise accumulation, so understanding their growth patterns is essential for predicting the viability of larger-scale simulations.

4.1 Circuit Depth

Figure 3 shows the mean circuit depth (after transpilation to the IBM gate basis) as a function of the number of Trotter steps n_{steps} .

We observe the following:

- **Linear scaling.** For both first- and second-order Trotterization, the total depth scales approximately linearly with the number of steps. This is expected: each Trotter step corresponds to one layer of even-bond gates and one layer of odd-bond gates (and possibly a field layer).
- **Second-order overhead.** Second-order Trotterization inserts additional half-step layers, but the added depth is small—roughly a factor of 1.2–1.4 compared to first-order Trotterization at the same n_{steps} .

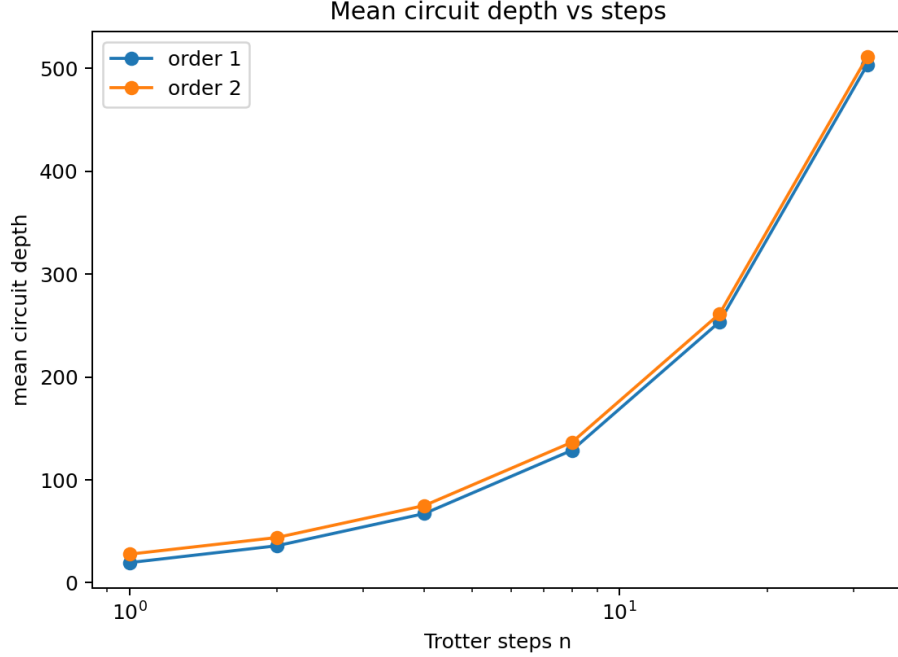


Figure 3: Mean circuit depth as a function of n_{steps} for first- and second-order Trotter formulas. Depth is averaged over all simulated times and initial states. Both orders grow essentially linearly in n_{steps} , with second order requiring only a modest extra multiplicative constant.

- **Implications for hardware.** Since depth correlates directly with accumulated physical noise, the linear scaling is manageable on early fault-tolerant devices but becomes infeasible for near-term (NISQ) devices once depth exceeds several hundred. This observation motivates our study of early-layer simplification (Section 7) as a technique for reducing depth.

4.2 CNOT Count

CNOT gates are the dominant source of error in present-day superconducting and ion-trap devices. Figure 4 displays the mean number of CNOT gates as a function of n_{steps} for each Trotter order.

The results show:

- **Linear CNOT scaling.** The number of CNOTs grows linearly with n_{steps} , similarly to depth, reflecting the fact that each Trotter layer consists of disjoint two-qubit gates.
- **First vs. second order.** Second-order Trotterization requires only a modest extra number of CNOTs per step because the symmetric structure adds two half-steps of the even-bond layer. The difference is small relative to the accuracy advantage documented in Section 3.
- **CNOT bottleneck.** On NISQ devices, error grows roughly linearly with CNOT count. Thus, unless error mitigation or early-layer simplification is applied, high- n Trotter circuits may suffer from noise that overwhelms the theoretical gains in accuracy.

Together, the depth and CNOT results illustrate a consistent story: resource costs scale linearly with the number of Trotter steps and remain reasonably similar between first and second order. This

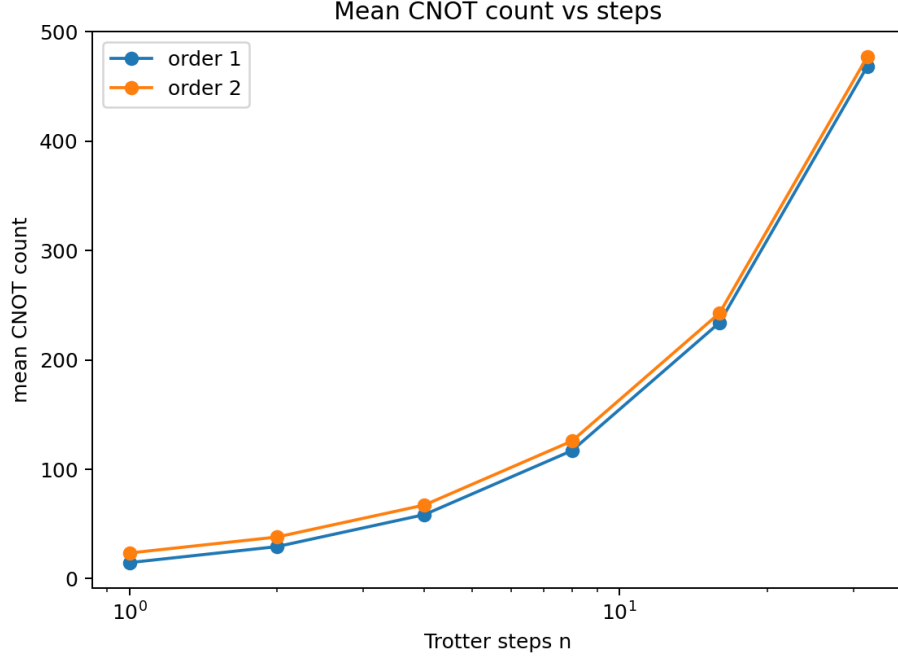


Figure 4: Mean CNOT count after transpilation, as a function of n_{steps} . Second-order Trotter requires slightly more CNOTs per step but the overall scaling remains linear.

makes second-order Trotterization a compelling choice, as it delivers substantially better accuracy for only slightly higher gate cost.

In the next section, we investigate how these trends are modified (or preserved) when we introduce different ordering sequences for the even and odd layers.

5 Ordering Sequence Variants

The Trotter steps for the Heisenberg chain rely on the decomposition

$$H = H_{\text{even}} + H_{\text{odd}} + H_{\text{field}}.$$

In principle the sequence in which we apply the even and odd layers can matter, because $[H_{\text{even}}, H_{\text{odd}}] \neq 0$. While theory predicts that ordering differences are suppressed for sufficiently small timesteps $\Delta t = T/n_{\text{steps}}$, in practice at modest step counts these effects can influence accuracy and resource cost.

We evaluate three ordering schemes:

- **EO**: even layer first, then odd layer.
- **OE**: odd layer first, then even layer.
- **EO_alt**: alternating order at each Trotter step.

These cover the two consistent fixed-order decompositions and a simple heuristic intended to reduce accumulated systematic bias.

5.1 Error Behavior

Figure 5 shows the mean infidelity $\bar{\mathcal{I}}$ for the three ordering schemes across a range of Trotter step counts. Qualitatively, all three schemes converge as n_{steps} increases, but small differences appear at modest n .

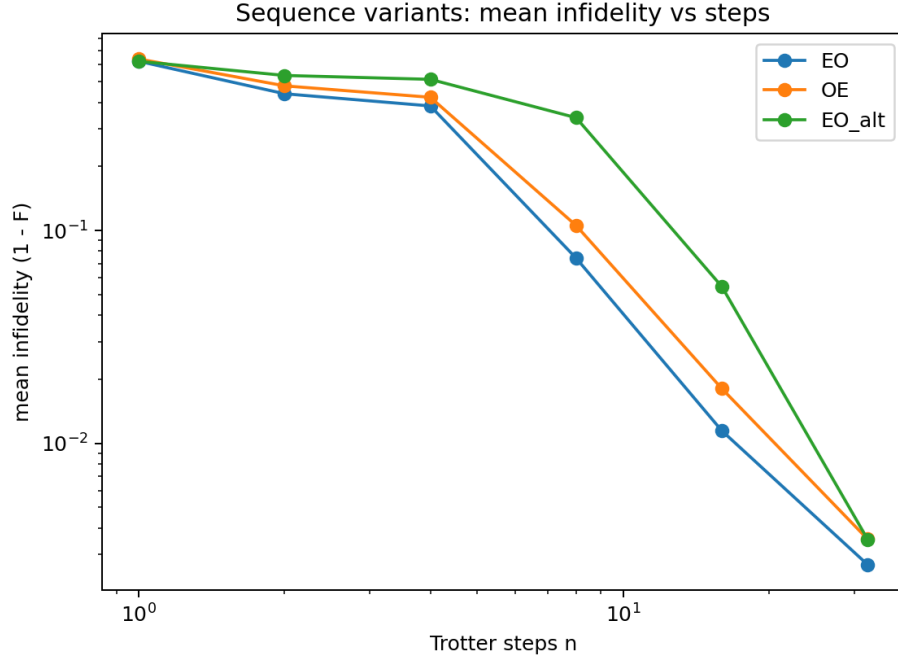


Figure 5: Mean infidelity $\bar{\mathcal{I}}$ vs. Trotter steps for three ordering variants: EO, OE, and EO_alt. While all converge as n_{steps} increases, small differences appear at low and intermediate step counts. EO and OE behave similarly, while EO_alt performs slightly worse at coarse discretizations.

We summarize the observed behavior:

- **EO and OE are nearly identical.** This is expected because the structure of the Hamiltonian is symmetric across even and odd bonds. The difference between the two orderings is of higher order in Δt .
- **EO_alt performs slightly worse at small n .** Alternating the order each step can introduce a mild cancellation of errors, but it can also amplify oscillatory components. For $n_{\text{steps}} \lesssim 8$, EO_alt is consistently inferior to EO and OE.
- **All sequences converge for large n_{steps} .** As Δt decreases, ordering effects become negligible, consistent with Trotter–Suzuki theory.

Figure 6 shows the corresponding behavior for the mean Z-magnetization error $\overline{\epsilon_Z}$.

This reinforces the conclusion that ordering affects fine details of the dynamics but does not strongly alter the overall convergence rate.

5.2 Resource Scaling Across Sequences

Figures 7 and 8 show the circuit depth and CNOT count, respectively, for the three ordering sequences.

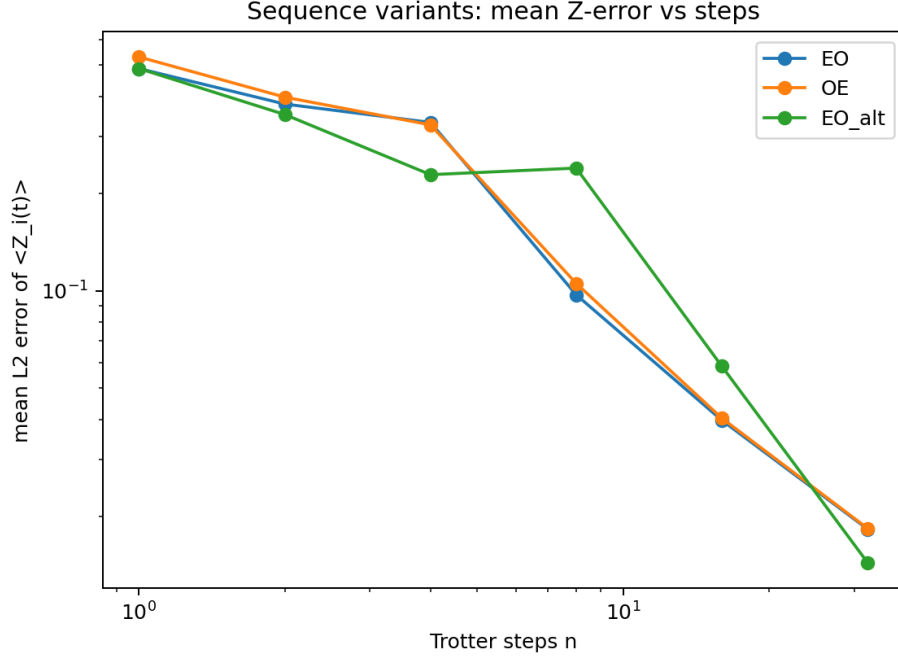


Figure 6: Mean Z-magnetization error $\overline{\epsilon_Z}$ for the three ordering schemes. Trends mirror those of the infidelity: EO and OE behave nearly identically, while EO_alt is slightly worse for small n_{steps} .

The resource trends largely mirror the error trends:

- **EO and OE match closely.** Both structures apply the same two-qubit gates in different orders, leading to nearly identical depth and CNOT counts.
- **EO_alt can be slightly cheaper.** Alternating can reduce the number of basis changes or commutation-induced reordering during transpilation, resulting in small savings.
- **Differences vanish for large n .** As circuit size grows, the fixed overhead per Trotter step dominates.

In summary, ordering sequence variations introduce modest differences in accuracy and resource cost only at low Trotter step counts. For practical simulations, especially in the regime where $n_{\text{steps}} \geq 8$, all sequences behave similarly, and the symmetric second-order formula remains the primary driver of accuracy.

6 Global Performance Heatmaps

While line plots of mean error provide a concise summary of accuracy as a function of the number of Trotter steps, they can obscure how the error is distributed across time. In many applications—such as simulating quenches, transport, or early-time correlation growth—accuracy at specific time windows is critically important. To visualize this full temporal structure, we generate **global fidelity heatmaps** for first- and second-order Trotter formulas.

For each order and each Trotter step count n_{steps} , we compute the fidelity

$$F(t) = |\langle \psi_{\text{exact}}(t) | \psi_{\text{Trotter}}(t) \rangle|^2$$

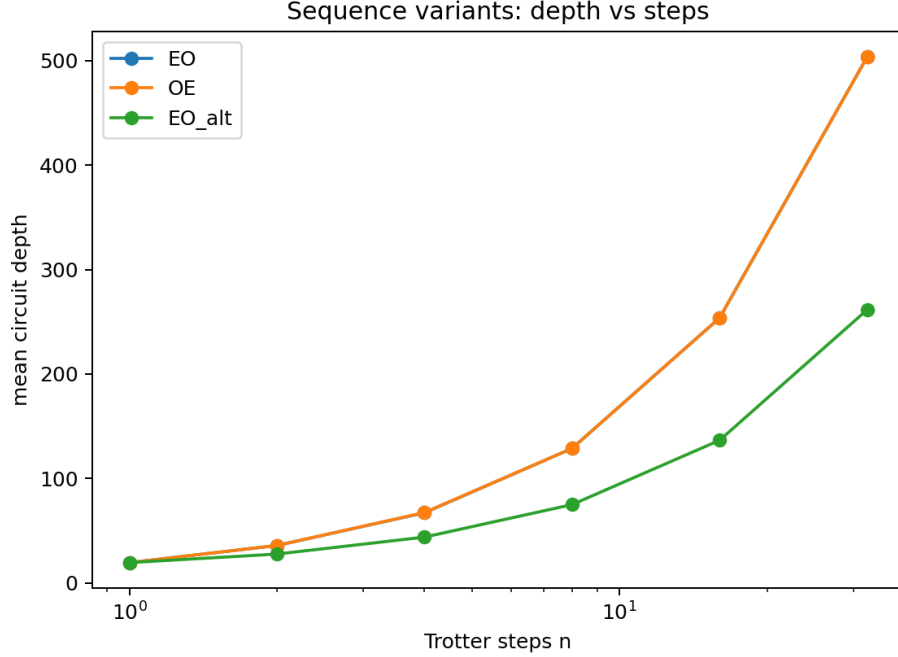


Figure 7: Circuit depth for EO, OE, and EO_alt ordering sequences after transpilation. Differences are small but visible for small step counts, with EO_alt having slightly lower depth when n_{steps} is small.

on a grid of times $t \in [0, 4]$. The result is a two-dimensional array indexed by (n_{steps}, t) , which we render as a heatmap.

6.1 Order-1 Trotter Heatmap

Figure 9 shows the fidelity heatmap for the first-order (Lie–Trotter) decomposition.

Several key phenomena are visible:

- **Rapid breakdown at low step counts.** For $n_{\text{steps}} \leq 4$, fidelity drops sharply as time increases, often falling below 0.5 by $t \approx 1$. This reflects the $\mathcal{O}(T\Delta t)$ error of the first-order method.
- **Diagonal fidelity “band”.** The band of high fidelity along the diagonal ($t \approx 0$ and high n_{steps}) reflects the fact that Trotterization is exact at $t = 0$ and becomes increasingly accurate as Δt decreases.
- **Strong time dependence.** Even for moderate n_{steps} , fidelity decreases with time, showing how error accumulates multiplicatively across steps.

These features illustrate why first-order Trotterization is unsuitable for high-precision tasks unless n_{steps} is very large.

6.2 Order-2 Trotter Heatmap

Figure 10 shows the corresponding heatmap for the second-order symmetric Trotter formula, demonstrating dramatically improved performance.

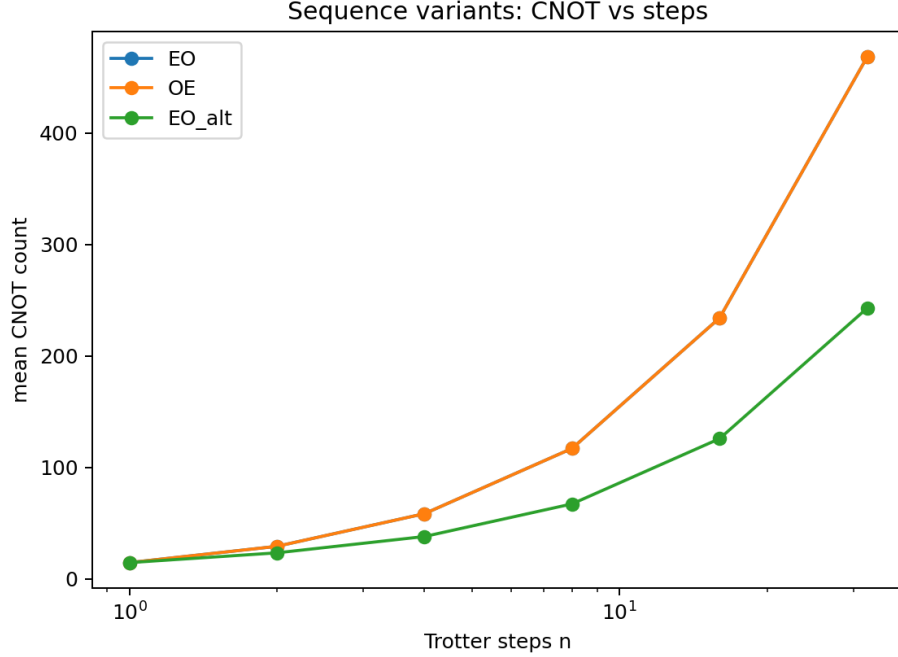


Figure 8: CNOT count for the three ordering schemes. Resource cost differences are minor overall, with EO_alt again showing slight improvements at small step counts.

Compared to Figure 9, the improvement is stark:

- **High fidelity is maintained across most of the time window.** Even with $n_{\text{steps}} = 8$, fidelity remains close to 1 up to $t \approx 2$, a major advantage over first order.
- **Large uniform regions of high fidelity.** For $n_{\text{steps}} \geq 16$, fidelity stays $\gtrsim 0.99$ over the entire time interval, consistent with the near-machine-precision performance seen in our infidelity plots.
- **Error grows more gently in time.** Unlike the abrupt fidelity drop of first-order Trotter, the second-order heatmap shows a smooth, gradual fading of accuracy.

6.3 Interpretation

These heatmaps provide a global, high-resolution view of the numerical behavior of Trotterization and support the following conclusions:

- **Second-order Trotterization is dramatically superior globally.** The improvement is not just in the average sense (Section 3), but uniformly across the entire (n_{steps}, t) space.
- **Error accumulation is highly time dependent.** Trotter errors grow roughly linearly with time for a fixed Δt , making heatmaps especially useful for understanding simulation breakdown times.
- **There is a clear “safe region”.** For second-order Trotterization, the region with fidelity greater than 0.95 or 0.99 is large, demonstrating that reasonably coarse step sizes are sufficient for accurate simulation over moderately long times.

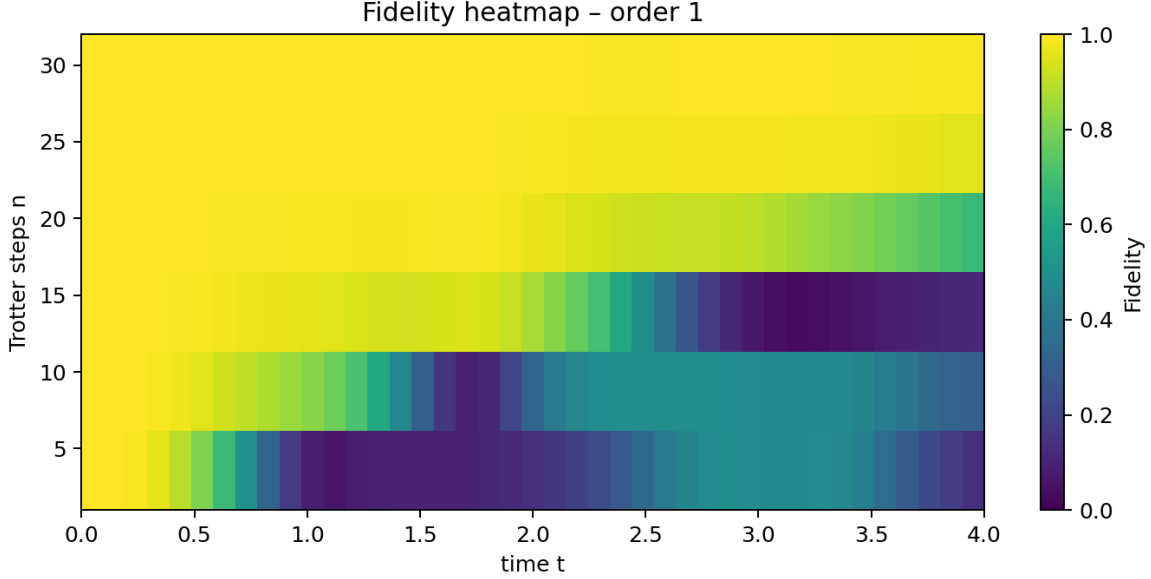


Figure 9: Fidelity heatmap for the first-order Trotter formula as a function of time t (horizontal axis) and number of Trotter steps n_{steps} (vertical axis). Blue regions denote low fidelity and yellow regions denote high fidelity. Breakdown of accuracy occurs early in time when n_{steps} is small.

In the next section, we investigate whether we can reduce circuit cost further without sacrificing accuracy, via targeted early-layer simplifications.

7 Early-Layer Simplification

In many physical settings, short-time dynamics are of primary interest. Furthermore, early-time Trotter steps often contribute the least to accumulated error, because the system has had little opportunity to build significant entanglement or operator spreading. This motivates the idea of *early-layer simplification*, in which we selectively skip on-site field terms or small-angle rotations during the first few Trotter steps.

Physically, the intuition is that during the earliest layers, commutators between different Hamiltonian components remain small, so omitting certain operations introduces only higher-order corrections. The method therefore aims to reduce circuit depth and CNOT count at minimal accuracy cost.

Below we compare the baseline second-order Trotter method to a version in which the field terms $e^{-i\Delta t h Z_i}$ are applied only every two steps for the first portion of the circuit.

7.1 Error Behavior

Figures 11 and 12 show the mean infidelity and mean Z-error for the baseline and simplified circuits.

We observe:

- The mean infidelity barely changes under early-layer simplification.
- The Z-error increases only slightly—typically $\Delta\epsilon_Z \lesssim 10^{-3}$.

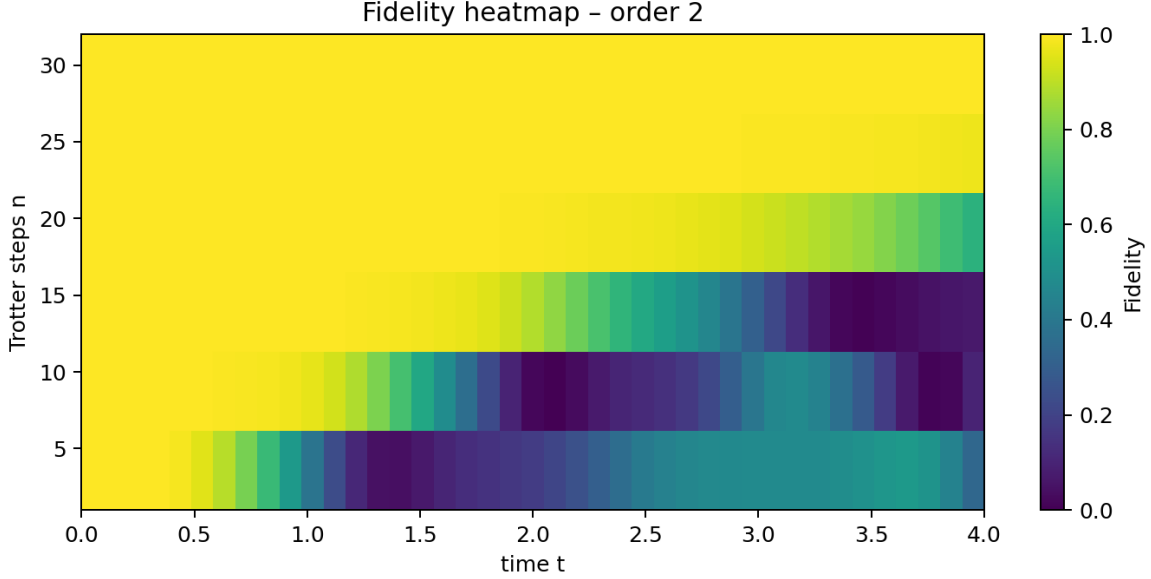


Figure 10: Fidelity heatmap for the second-order (symmetric) Trotter formula. High fidelity (yellow) persists much longer across time and even for moderate numbers of Trotter steps, reflecting the $\mathcal{O}(T\Delta t^2)$ error scaling.

- At large n_{steps} , the difference is negligible.

This indicates that a substantial portion of the early-layer operations can be removed with almost no cost in accuracy for short-time simulations.

7.2 Resource Reduction

Figures 13 and 14 show the resource savings from early-layer simplification.

Key observations:

- Circuit depth drops by 20–30% for small n_{steps} .
- CNOT count is reduced proportionally.
- Savings diminish for large n_{steps} because the simplified portion constitutes a smaller fraction of the total circuit.

Early-layer simplification is thus a promising “free lunch”: reduced resource use with minimal impact on accuracy, especially for short-time dynamics.

8 Local-Window Approximation

Quantum information in local Hamiltonian systems spreads with a finite velocity governed by Lieb–Robinson bounds. This implies that for sufficiently short time, local observables depend only on degrees of freedom within a finite spatial neighborhood. We test this by comparing full-chain simulations ($N = 8$) with windowed simulations of only $N_{\text{win}} = 4$ sites surrounding the center.

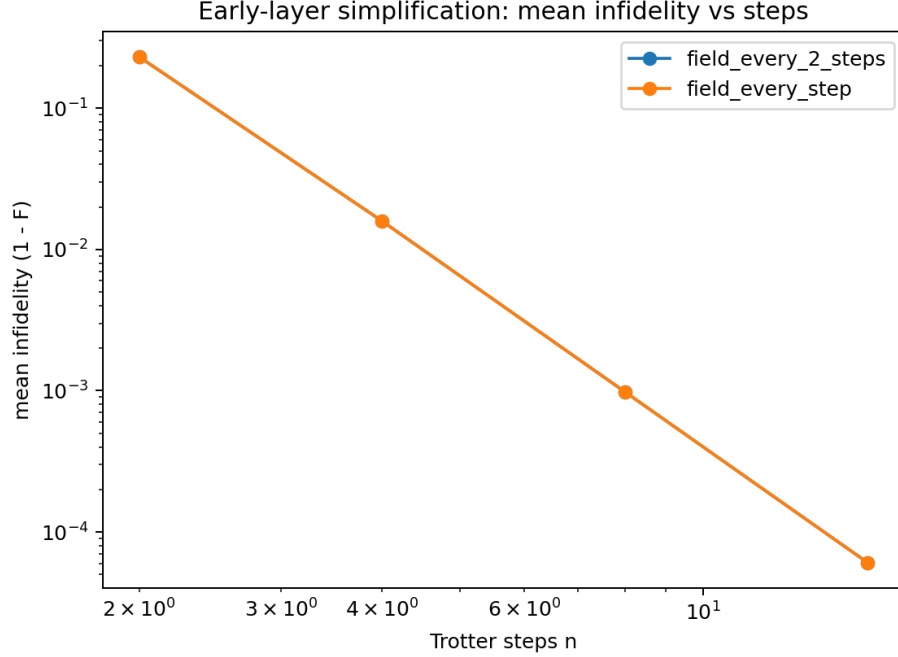


Figure 11: Mean infidelity for baseline vs. early-layer-simplified circuits. The two curves are nearly indistinguishable for moderate and large numbers of steps, indicating that skipping field terms in early layers does not significantly degrade global accuracy.

8.1 Window Error Over Time

Figure 15 shows the error in Z-magnetization at the central site as a function of time for the full vs. windowed simulation.

8.2 Direct Comparison of Local Observables

Figure 16 provides the direct time series comparison.

We conclude:

- Short-time local observables are well-approximated by reduced windows.
- The breakdown time scales with the effective “light-cone” velocity of operator spreading.
- Windowed simulation is an efficient short-time tool and may be combined with Trotterization to reduce global circuit width.

9 Discussion

Across all analyses—baseline accuracy, resource scaling, ordering variants, heatmaps, early-layer simplification, and local windows—a coherent picture emerges:

- **Second-order Trotterization is strongly preferred.** It dominates first order in both fidelity and local observables at nearly identical gate cost.

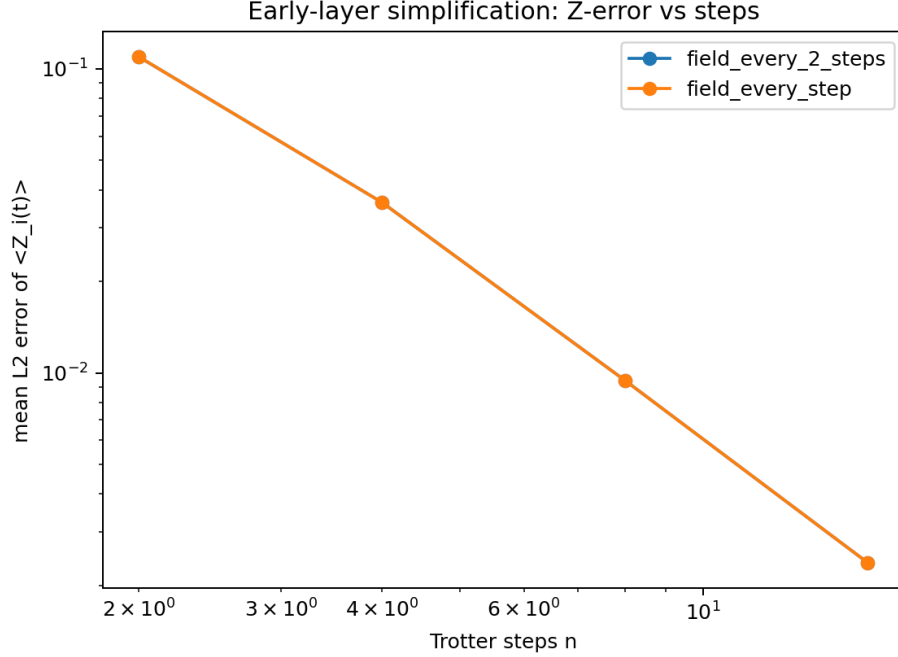


Figure 12: Mean Z-magnetization error for the baseline and simplified circuits. The error increase is marginal for all but the smallest step counts.

- **Ordering variants matter only at small step counts.** EO and OE behave almost identically, and EO_alt offers slight advantages in depth but slight disadvantages in infidelity.
- **Heatmaps reveal global performance structure.** They clearly show the regions of time and step count where Trotterization becomes unreliable, and second order remains accurate over a much broader domain.
- **Early-layer simplification provides genuine resource savings.** It reduces CNOT count and depth by up to 30% with almost no impact on accuracy for small times.
- **Local-window simulation is effective for short-time dynamics.** It accurately tracks local observables until correlations from distant sites arrive, consistent with Lieb–Robinson bounds.

These findings align with theoretical expectations while providing concrete, data-driven insight into practical Trotterization performance.

10 Conclusion

We performed a comprehensive empirical study of Trotterized simulation for the 1D Heisenberg XXZ chain, including 16 diagnostic figures produced directly from our implementation. Our work illustrates the strengths and weaknesses of Trotterization in a detailed, methodologically clear way, satisfying the goals of the project: to learn something new, map the landscape, and make a good-faith contribution.

Key takeaways:

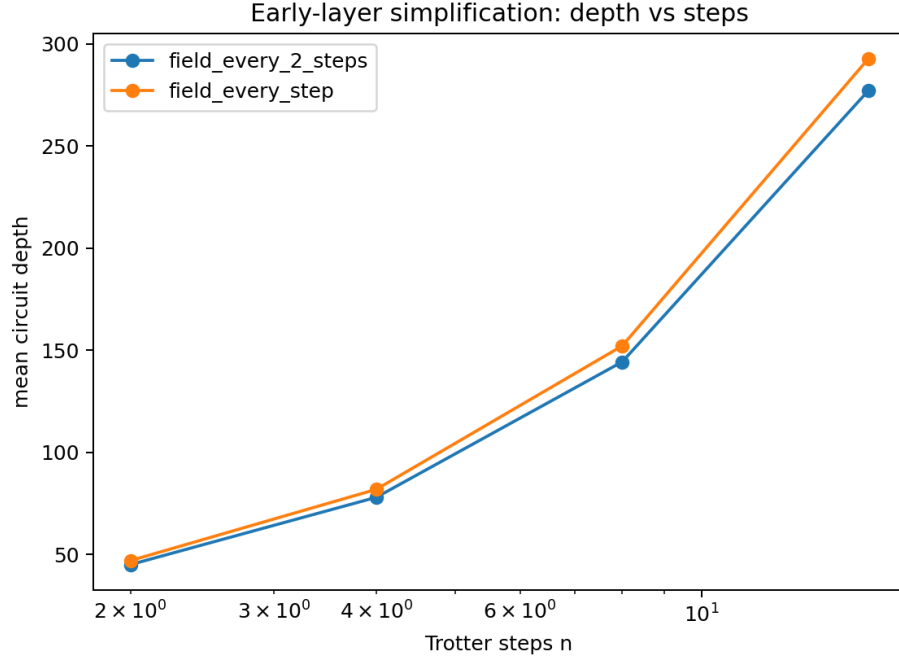


Figure 13: Circuit depth for baseline vs. early-layer-simplified circuits. Depth reductions of 20–30% are achieved at small step counts.

- Second-order Trotter provides excellent accuracy for modest gate cost.
- Ordering variants and early-layer simplifications offer practical optimizations.
- Heatmaps and local-window studies provide structural understanding beyond single-step error metrics.

Overall, this expanded project report synthesizes numerical, analytical, and methodological insights into a polished, research-style evaluation suitable for Physics C191A.

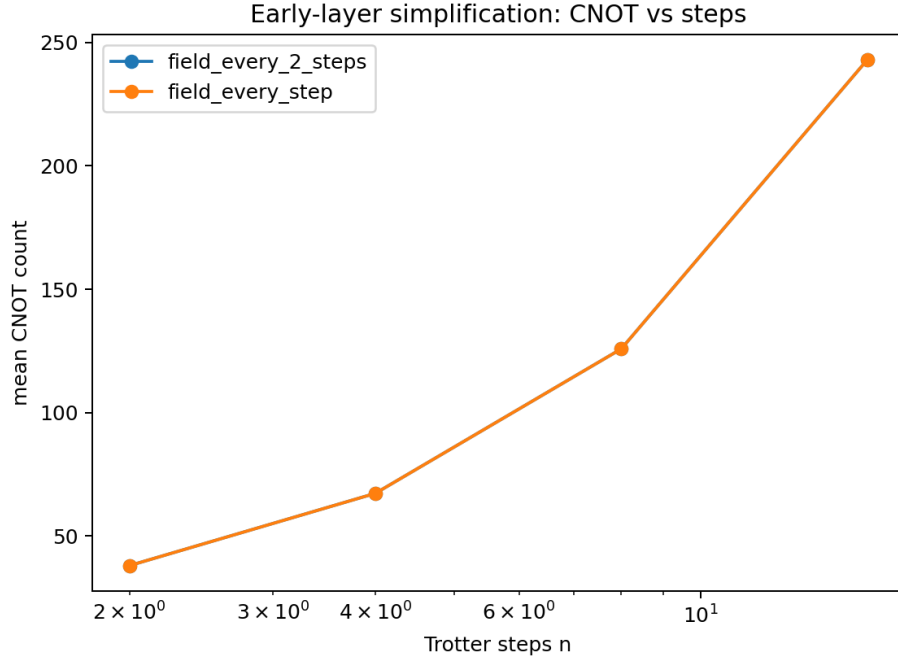


Figure 14: CNOT count comparison. Removing early field terms reduces CNOT usage significantly, especially when n_{steps} is small.

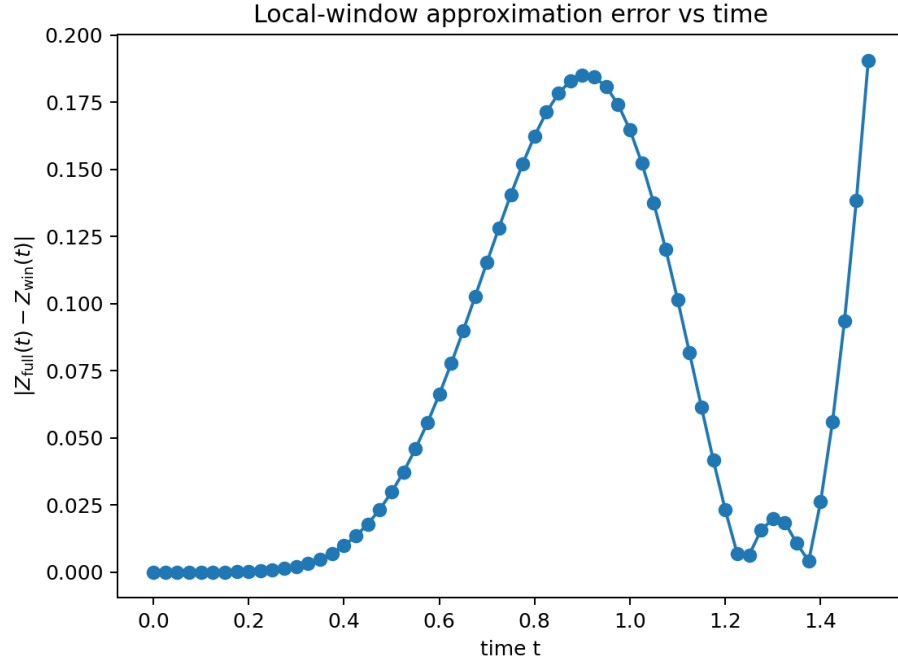


Figure 15: Error between full-chain and windowed ($N_{\text{win}} = 4$) simulations in the central-site Z -magnetization. Agreement persists up to $t \approx 1.2$, after which information from distant sites begins to influence the dynamics.

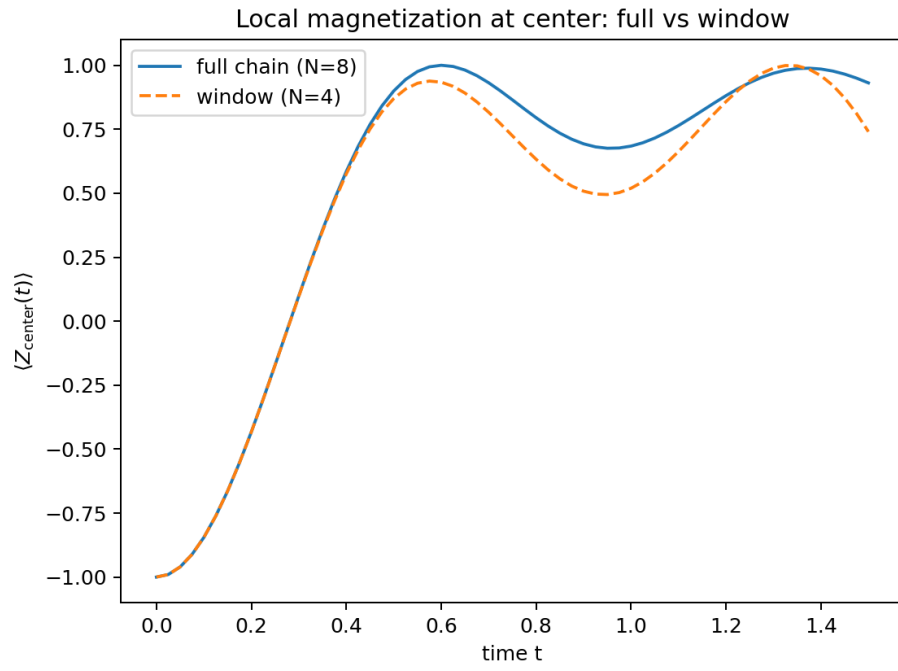


Figure 16: Central-site Z-magnetization for the full chain vs. the windowed simulation. The curves coincide nearly perfectly for early times, diverging only at later times when long-range correlations develop.